Chapter 3

Extensive Form Games

In this chapter, we study *extensive form representation* of a game, which provides a detailed representation of a game, including the sequence of moves and information accrual to players. We explain the important notions underlying extensive form representation. In this book, we mostly deal with strategic form representation; in this chapter, we bring out the connection between extensive form and strategic form representations. We show how any extensive form game can be transformed into a strategic form game.

The extensive form representation of a game provides a detailed and richly structured way to describe a game. This form was first proposed by von Neumann and Morgenstern [1] and was later refined by Kuhn [2]. The extensive form captures complete sequential play of a game. Specifically it captures (1) who makes a move at any given time (2) what actions each player may play (3) what the players know before playing at each stage (4) what the outcomes are as a function of the actions, and (5) payoffs that players obtain from each outcome. Extensive form games with a finite number of players and with a finite number of actions available to each player are depicted graphically using game trees.

3.1 Illustrative Examples

We first present several examples before we formally define an extensive form game.

Example 3.1 (Matching Pennies with Observation). In the matching pennies game, there are two players, 1 and 2, each of whom has a rupee coin. One of the players puts down his rupee coin heads or tails up. The other player sees the outcome and puts down her coin heads up or tails up. If both the coins show heads or both the coins show tails, player 2 gives one rupee to player 1 who thus becomes richer by one rupee. If one of the coins shows heads and the other coin shows tails, then player 1 pays one rupee to player 2 who becomes richer by one rupee. Depending on whether player 1 or player 2 moves first, there are two versions of this game. Figure 3.1 shows the game tree when player 1 moves first while Figure 3.2 shows the game tree when player 2 moves first. In the game tree representation, the nodes are of three types: (1) *root node* (initial decision node); (2)

internal nodes (which are decision nodes); and (3) *leaf nodes* or *terminal nodes* (which are outcome nodes). Each possible sequence of events that could occur in the game is captured by a path of links from the root node to one of the terminal nodes. When the game is played, the path that represents the sequence of events is called the *path of play*. Each decision node is labeled with the player who takes a decision at that node. Also note that each decision node can be uniquely identified by a sequence of actions leading to that decision node from the root node. The links that are outgoing at the decision node are labeled with the actions the player may select at that node. Note that each node represents not only the current position in the game but also how it was reached. The terminal nodes are labeled with the payoffs that the players would get in the outcomes corresponding to those nodes. □

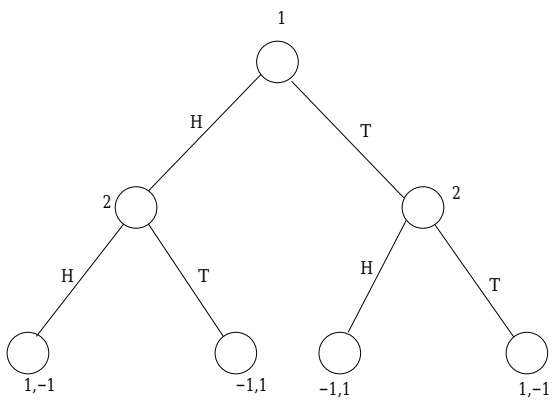

Fig. 3.1: Matching pennies game with observation when player 1 moves first

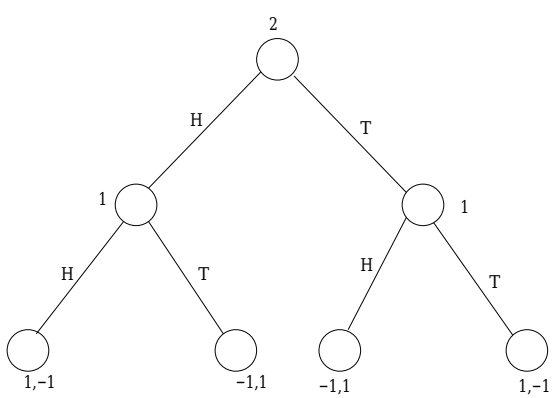

Fig. 3.2: Matching pennies game with observation when player 2 moves first

Example 3.2 (Matching Pennies without Observation). In this case, one of the players places his rupee coin heads up or tails up. The other player *does not observe the outcome* and only puts down her rupee coin heads up or tails up. Depending on whether player 1 moves first or player 2 moves first, we obtain the game tree of Figure 3.3 or Figure 3.4, respectively. Note that the game trees of Figures 3.1 and 3.3 are virtually the same except that the two decision nodes corresponding to player 2 in Figure 3.3 are connected with dotted lines. Similarly the game trees of Figures 3.2 and 3.4 are the same except that the two decision nodes corresponding to player 1 in Figure 3.4 are connected with dotted lines. A set of nodes that are connected with dotted lines is called an *information set*. When the game reaches a decision node in an information set, the player involved at that node does not know the node in the information set she is in. The reason for this is that the player cannot observe the previous moves in the game. □

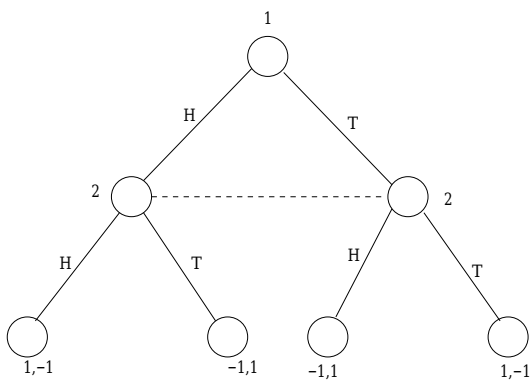

Fig. 3.3: Matching pennies game without observation when player 1 moves first

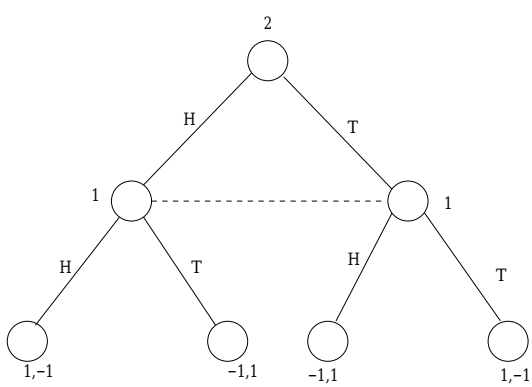

Fig. 3.4: Matching pennies game without observation when player 2 moves first

Example 3.3 (Matching Pennies with Simultaneous Play). In this version of the game, the two players put down their rupee coins simultaneously. Clearly, each player has no opportunity to observe the outcome of the move of the other player. The order of play is obviously irrelevant here. Thus both the game trees depicted in Figure 3.3 and Figure 3.4 provide a valid representation of this version of the game. □

3.2 Extensive Form Games: Definitions

We now formally define an extensive form game. This definition follows closely the one given by Osborne [3]. First we define an information set.

Definition 3.1 (Information Set). *An information set of a player is a set of that player's decision nodes that are indistinguishable to her.*

An information set of a player describes a collection of all possible distinguishable circumstances in which the player is called upon to make a move. Since each decision node corresponds uniquely to a sequence of actions from the root node to the decision node, each information set of a player consists of all proper subhistories relevant to that player which are indistinguishable to that player. Clearly, in every node within a given information set, the corresponding player must have the same set of possible actions.

Example 3.4. In the matching pennies game shown in Figure 3.3, the only information set of a player 1 is the singleton $\{\varepsilon\}$ consisting of the empty history. The information set of player 2 is the set $\{H, T\}$ that consists of the proper histories H and T which are indistinguishable to player 2. On the other hand, in the game shown in Figure 3.1, player 1 has only one information set namely $\{\varepsilon\}$ whereas player 2 has two information sets $\{H\}$ and $\{T\}$ because these two proper subhistories are distinguishable to player 2. □

Definition 3.2 (Extensive Form Game). *An extensive form game* Γ *consists of a tuple* $\Gamma = \langle N, (A_i)_{i\in N}, \mathbb{H}, P, (\mathbb{I}_i)_{i\in N}, (u_i)_{i\in N}\rangle$ *where*

- $N = \{1, 2, \ldots, n\}$ *is a finite set of players*
- A_i *for* $i = 1, 2, \ldots, n$ *is the set of actions available to player* i *(action set of player* i*)*
- $\mathbb{H}$ *is the set of all* terminal histories *where a terminal history is a path of actions from the root to a terminal node such that it is not a proper subhistory of any other terminal history. Denote by* $S_{\mathbb{H}}$ *the set of all proper subhistories (including the empty history* ε*) of all terminal histories.*
- $P : S_{\mathbb{H}} \to N$ *is a player function that associates each proper subhistory to a certain player*
- $\mathbb{I}_i$ *for* $i = 1, 2, \ldots, n$ *is the set of all information sets of player* i
- $u_i : \mathbb{H} \to \mathbb{R}$ *for* $i = 1, 2, \ldots, n$ *gives the utility of player* i *corresponding to each terminal history.*

Example 3.5. We illustrate the above definition for the matching pennies game shown in Figure 3.1.

$$N = \{1, 2\}$$

$$A_1 = A_2 = \{H, T\}$$

$$\mathbb{H} = \{(H, H), (H, T), (T, H), (T, T)\}$$

$$S_{\mathbb{H}} = \{\varepsilon, H, T\}$$

$$P(\varepsilon) = 1;\ P(H) = 2;\ P(T) = 2$$

$$\mathbb{I}_1 = \{\{\varepsilon\}\};\ \ \mathbb{I}_2 = \{\{H\}, \{T\}\}$$

$$u_1(HH) = 1;\ u_1(HT) = -1;\ u_1(TH) = -1;\ u_1(TT) = 1$$

$$u_2(HH) = -1;\ u_2(HT) = 1;\ u_2(TH) = 1;\ u_2(TT) = -1$$

It is clear that action sets of different players can be deduced from the terminal histories and the player function. □

Note. Though the action sets of players can be deduced from terminal histories and the player function, we explicitly include action sets as a part of definition of an extensive form game for ease of understanding.

Example 3.6 (Entry Game). In this game, there are two players, 1 and 2. Player 1 is called *challenger* and player 2 is called *incumbent*. Figure 3.5 shows the game tree. Player

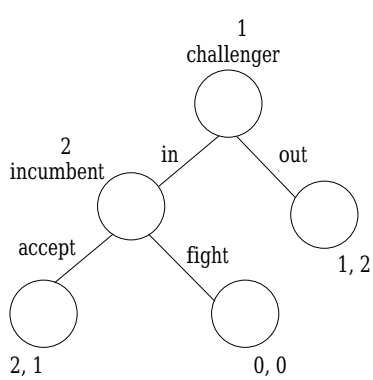

Fig. 3.5: Game tree for entry game

1 (challenger) either decides to challenge the incumbent (action: in) or drops out (action: out). Player 2 (incumbent) either decides to fight or accommodate the challenger in case the challenger decides to confront the incumbent. The respective payoffs are shown in the

game tree. For this game, we have

$$
\begin{aligned}
&N = \{1, 2\}; \quad A_1 = \{\text{in, out}\}; \quad A_2 = \{\text{accept, fight}\} \\
&\mathbb{H} = \{(\text{in, accept}), (\text{in, fight}), (\text{out})\} \\
&S_{\mathbb{H}} = \{\epsilon, (\text{in})\} \\
&P(\epsilon) = 1 \\
&P(\text{in}) = 2 \\
&\mathbb{I}_1 = \{\{\varepsilon\}\}; \ \ \mathbb{I}_2 = \{\{\text{in}\}\} \\
&u_1(\text{in, accept}) = 2; \quad u_1(\text{in, fight}) = 0; \quad u_1(\text{out}) = 1 \\
&u_2(\text{in, accept}) = 1; \quad u_2(\text{in, fight}) = 0; \quad u_2(\text{out}) = 2
\end{aligned}
$$

Note again that the action sets A_1 and A_2 can be deduced from the terminal histories and the player function. □

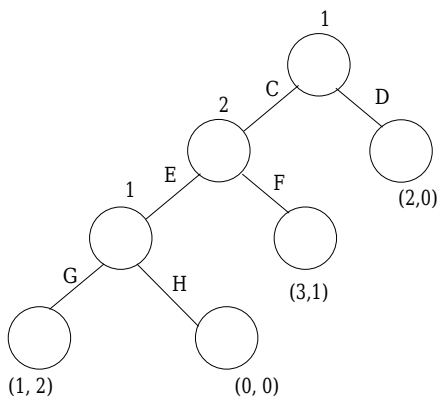

Fig. 3.6: Another game tree

Example 3.7. Consider the game tree shown in Figure 3.6. For this game, it is easy to see that

$$N = \{1, 2\}; \quad A_1 = \{C, D, G, H\}; \quad A_2 = \{E, F\}$$

The terminal histories are given by

$$\mathbb{H} = \{(C, E, G), (C, E, H), (C, F), (D)\}$$

The proper subhistories of terminal histories are given by

$$S_{\mathbb{H}} = \{\epsilon, (C), (C, E)\}$$

The player function is given by

$$P(\epsilon) = 1; \quad P(C) = 2; \quad P(C, E) = 1$$

The information sets are given by

$$\mathbb{I}_1 = \{\{\varepsilon\}, \{(C, E)\}\}; \;\; \mathbb{I}_2 = \{\{(C)\}\}$$

The utility functions are given by

$$\begin{aligned}
&u_1(C, E, G) = 1; \quad u_1(C, E, H) = 0;\\
&u_1(C, F) = 3; \quad u_1(D) = 2;\\
&u_2(C, E, G) = 2; \quad u_2(C, E, H) = 0;\\
&u_2(C, F) = 1; \quad u_2(D) = 0.
\end{aligned}$$

This completes the description of the game shown in Figure 3.6. □

Definition 3.3 (Perfect Information and Imperfect Information Games). *An extensive form game with perfect information is one in which all the information sets are singletons. If at least one information set of at least one player has two or more elements, the game is said to be of imperfect information.*

In a game with perfect information, each player is able to observe all previous moves or the entire history thus far. Each player knows precisely where she is currently and also knows precisely how she has reached that node.

Example 3.8. As immediate examples, the games depicted in Figures 3.1, 3.2, 3.5, and 3.6 are games with perfect information while the games shown in Figures 3.3 and 3.4 are games with imperfect information. The matching pennies game with simultaneous play is obviously a game with imperfect information. □

3.3 Transforming Extensive Form to Strategic Form

The notion of a strategy is one of the most important notions in game theory. A *strategy* can be described as a complete action plan that specifies what a player will do at each of the information sets where he is called upon to play.

Recall that $\mathbb{I}_i$ denotes the set of all information sets of player i in the given game. Let A_i as usual denote the actions available to player i. Given an information set $J \in \mathbb{I}_i$, let $C(J) \subseteq A_i$ be the set of all actions possible to player i in the information set J. Then we define a strategy of a player formally as follows.

Definition 3.4 (Strategy). *A strategy s_i of player i is a mapping $s_i : \mathbb{I}_i \to A_i$ such that $s_i(J) \in C(J)\ \forall J \in \mathbb{I}_i$.*

The strategy s_i for player i is a complete contingent plan that specifies an action for every information set of the player. A strategy thus determines the action the player will choose in every stage or history of the game the player is called upon to play. In fact, the player can prepare a look-up table with two columns, one

for her information sets and the other for corresponding actions; the player or a representative of the player can then take over and play the game using table look-up. Different strategies of the player correspond to different contingent plans of actions. We illustrate the notion of strategy through an example.

Example 3.9 (Strategies in Matching Pennies with Observation). Consider the game shown in Figure 3.1. We have $\mathbb{I}_1 = \{\{\varepsilon\}\}$; $\mathbb{I}_2 = \{\{H\}, \{T\}\}$. Player 1 has two strategies:

$$s_{11} : \{\varepsilon\} \to H$$

$$s_{12} : \{\varepsilon\} \to T$$

Player 2 has the following four strategies:

$$s_{21} : \{H\} \to H; \quad \{T\} \to H$$

$$s_{22} : \{H\} \to H; \quad \{T\} \to T$$

$$s_{23} : \{H\} \to T; \quad \{T\} \to H$$

$$s_{24} : \{H\} \to T; \quad \{T\} \to T$$

The payoffs obtained by the players 1 and 2 can now be described by Table 3.1. Note, for example, that when the strategy of player 1 is s_{11}, the player plays H and when the strategy of player 2 is s_{21}, the player 2 plays H, leading to the payoffs 1, -1.

	2			
1	s_{21}	s_{22}	s_{23}	s_{24}
s_{11}	$1, -1$	$1, -1$	$-1, 1$	$-1, 1$
s_{12}	$-1, 1$	$1, -1$	$-1, 1$	$1, -1$

Table 3.1: Payoffs obtained in matching pennies with observation

The above game is a *strategic form game* equivalent of the original extensive form game. For the game shown in Figure 3.2, player 2 will have two strategies and player 1 will have four strategies and a payoff matrix such as above can be easily derived. □

Example 3.10 (Strategies in Matching Pennies without Observation). Consider the game shown in Figure 3.3. It is easy to see that $\mathbb{I}_1 = \{\{\varepsilon\}\}$ and $\mathbb{I}_2 = \{\{H, T\}\}$. Here player 1 has two strategies and player 2 has two strategies as shown below.

$$s_{11} : \{\varepsilon\} \to H$$

$$s_{12} : \{\varepsilon\} \to T$$

$$s_{21} : \{H, T\} \to H$$

$$s_{22} : \{H, T\} \to T$$

The payoff matrix corresponding to all possible strategies that can be played by the players is shown in Table 3.2.

	2	
1	s_{21}	s_{22}
s_{11}	$1, -1$	$-1, 1$
s_{12}	$-1, 1$	$1, -1$

Table 3.2: Payoffs obtained in matching pennies without observation

Clearly, the matching pennies game with simultaneous moves also will have the same strategies and payoff matrix as above. □

Example 3.11. Consider the game in Figure 3.6. Player 1 has four strategies given by

$$s_{11} : \{\varepsilon\} \to C; \quad \{(C, E)\} \to G$$

$$s_{12} : \{\varepsilon\} \to C; \quad \{(C, E)\} \to H$$

$$s_{13} : \{\varepsilon\} \to D; \quad \{(C, E)\} \to G$$

$$s_{14} : \{\varepsilon\} \to D; \quad \{(C, E)\} \to H$$

For the sake of convenience, let us denote the above strategies by CG, CH, DG, and DH, respectively. Player 2 has two strategies given by

$$s_{21} : \{C\} \to E$$

$$s_{21} : \{C\} \to F$$

For the sake of convenience, let us denote the above strategies by E and F, respectively. If S_1 and S_2 are the sets of strategies of players 1 and 2 respectively, it can be seen that

$$S_1 = \{CG, CH, DG, DH\}$$
$$S_2 = \{E, F\}$$

The set of strategy profiles, $S_1 \times S_2$, is given by

$$S_1 \times S_2 = \{(CG, E), (CG, F), (CH, E), (CH, F), (DG, E), (DG, F), (DH, E), (DH, F)\}$$

Note that a strategy profile uniquely determines a terminal history. For example, the profile (CG, E) corresponds to the terminal history (C, E, G); the profile (CG, F) corresponds to the terminal history (C, F); the profiles (DH, E) as well as (DH, F) correspond to the terminal history (D), etc. This example motivates the following definition. □

Definition 3.5 (Outcome). *Given an extensive form game Γ and a strategy profile $s = (s_1, \ldots, s_n)$ in the game, the outcome resulting out of the terminal history corresponding to the strategy profile s is called the outcome of s and is denoted by $O(s)$.*

Note. It is to be noted that every extensive form game has a unique strategic form representation. The uniqueness is up to renaming or renumbering of strategies. We can also immediately observe that a given strategic form game may correspond to multiple extensive form games.

Note. We have seen that any given extensive form game has an equivalent strategic form game. However, the equivalent strategic form representation may or may not contain all of the strategically relevant information present in the extensive form representation. In fact, the strategic form representation suppresses the dynamics of the game because of simultaneous play. In this book, we mostly focus on the strategic form representation. It is to be noted that dynamic games where there is sequential play as well as information accrual to the players during the play of the game warrant extensive form representation.

3.4 Summary and References

Following is a summary of salient points that we have covered in this chapter.

- Extensive form games provide a detailed representation of sequence of play and information accrual by players in a game. Finite extensive form games can be represented using game trees which consist of decision nodes and terminal nodes. Each decision node corresponds to a certain player and the player is required to choose an action in the decision node.
- An important notion in an extensive form games is that of an *information set* of a player. An information set of a player is a set of decision nodes of the player that are indistinguishable to the player (the player does not know in which of these decision nodes she is in).
- An extensive form game with perfect information is one in which all information sets of all players are singletons. This implies that at every decision node, the corresponding player has knowledge of the entire history until reaching that decision node. An extensive form game with imperfect information is one where at least one information set of at least one player is not a singleton.
- An extensive form game can be transformed into an equivalent strategic form game using the notion of a strategy. A strategy of a player is a complete action plan that specifies which action the player will choose in each of her information sets.
- A strategic form game often suppresses the dynamics of the game. However, it simplifies the analysis of games and it suffices to work with the strategic form representation for finding answers to many useful analysis questions.
- A given strategic form game could correspond to multiple extensive form games while a given extensive form game when transformed into strategic form yields a representation that is unique in structure.

Much of the material in this chapter is based on relevant discussions in the books by Osborne [3] and by Mas-Colell, Whinston, and Green [4]. In this book, we will be dealing mostly with strategic form games. In Chapter 6, we briefly return to extensive form games to introduce the notion of subgame perfect equilibrium. For a detailed treatment of extensive form games, we refer the reader to the books by Osborne [3], Myerson [5], and Maschler, Solan, and Zamir [6].

References

[1] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior.* Princeton University Press, 1944.

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[3] Martin J. Osborne. *An Introduction to Game Theory.* The MIT Press, 2003.

[4] Andreu Mas-Colell, Michael D. Whinston, and Jerry R. Green. *Microeconomic Theory.* Oxford University Press, 1995.

[5] Roger B. Myerson. *Game Theory: Analysis of Conflict.* Harvard University Press, Cambridge, Massachusetts, USA, 1997.

[6] Michael Maschler, Eilon Solan, and Shmuel Zamir. *Game Theory.* Cambridge University Press, 2013.

3.5 Exercises

(1) You might know the tic-tac-toe game. Sketch a game tree for this game.

(2) In a game, a certain player has m information sets indexed by $j = 1, 2, \ldots, m$. There are k_j possible actions for information set j. How many strategies does the player have?

(3) For game shown in Figure 3.7, write down the terminal histories, proper subhistories, information sets, and the strategic form representation.

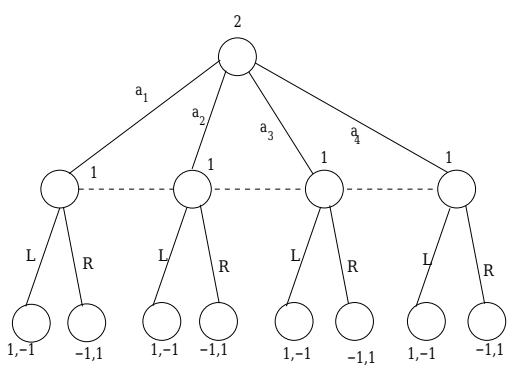

Fig. 3.7: An extensive form game